

1 Prompt Outlines

This appendix summarizes the prompt outlines used in LMVQA. The outlines abstract the implemented prompts by showing their roles, inputs, grounding rules, and expected output formats.

1.1 Main LMVQA Prompt Outlines

(a) Prompt Outline for Non-Diagram Frame Captioning

- (I) Role: an assistant analyzing a video frame from an educational lecture or tutorial.
- (II) Frame metadata: {timestamp}.
- (III) Input: a sampled video frame image.
- (IV) Task: describe visible slide, screen, chart, or non-diagram content; extract and summarize visible text when useful.
- (V) Grounding rules: ignore irrelevant UI elements, avoid redundancy, and describe only visible content.
- (VI) Output: a fluent structured English frame description.

(b) Prompt Outline for Diagram-Like Frame Detection

- (I) Role: a fast triage classifier.
- (II) Input: a single video frame image.
- (III) Task: decide whether the frame is diagram-like. Diagram-like visuals include diagrams, flowcharts, schematics, architecture diagrams, charts, tables, UI wireframes, whiteboards with boxes or arrows, and equations.
- (IV) Output: LABEL: <DIAGRAM|NOT_DIAGRAM>; TYPES: <types>; CONF: <0.00-1.00>.

(c) Prompt Outline for Seven-Way Diagram Classification

- (I) Role: a fast diagram-type classifier.
- (II) Input condition: the image is already known to be diagram-like.
- (III) Categories: UML_CLASS, UML_SEQUENCE, UML_STATE_ACTIVITY, UML_USE_CASE, NETWORK_TOPOLOGY, ARCH_WORKFLOW, or OTHER.
- (IV) Output: CATEGORY: <category>; CONF: <0.00-1.00>.

(d) Seven Diagram Categories Used for Category-Specific Extraction

UML_CLASS: diagram title; scope or packages; classes and interfaces; relationships; key takeaways.
UML_SEQUENCE: diagram title; participants or lifelines; temporally ordered messages; activations; combined fragments; key scenario summary.
UML_STATE_ACTIVITY: diagram title; subtype; nodes; transitions or flows; composite states, swimlanes, forks, joins, and behavior summary.
UML_USE_CASE: diagram title; system boundary; actors; use cases; associations; include, extend, and generalization relationships.
NETWORK_TOPOLOGY: title or legend; nodes and endpoints; links; traffic or flows; key topology takeaway.
ARCH_WORKFLOW: title; components or modules; interfaces or artifacts; connections or flows; workflow steps.
OTHER: title; diagram type guess; main elements; relationships or structure; key takeaway.

(e) Prompt Outline for Diagram Frame Extraction

- (I) Role: a diagram-frame describer specialized according to the selected diagram category.
- (II) Frame metadata: {timestamp}.
- (III) Input: the diagram-like video frame image.
- (IV) Generic schema: title, purpose, global layout, nodes, edges, flow sequence, text and labels, data or variables, context anchors, and key takeaway.
- (V) Category-specific schema: refined according to the seven-way classifier output.
- (VI) Grounding rules: describe only visible content, quote labels when possible, mark unclear text as [illegible], and do not invent facts.
- (VII) Output: exhaustive structured plain English text, not JSON.

(f) Prompt Outline for RAG-Based Video Question Answering

- (I) Role: an assistant answering questions about lecture videos using timestamped multimodal information.
- (II) Video metadata: {video_duration}.
- (III) Retrieved context: {retrieved_context}, formatted as [source] <timestamp>: <text>.
- (IV) User question: {user_question}.
- (V) Grounding rules: answer only from retrieved segments; do not introduce unsupported information; use the insufficient-information answer when needed.
- (VI) Output: Answer: plus final answer, and Used_context: with 1–10 relevant timestamped segments.

Fig. 1: Outlines of the main prompts used in LMVQA.